CN Lab 1: Understanding the Basic Network Equipment

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Title: Understanding the Basic Network Equipment

Theory

a. Network Equipment

Network equipment refers to the physical devices used to connect computers and other devices to form a network, enabling communication and data transfer between them. These devices are essential for building both small local area networks (LANs) and large wide area networks (WANs).

b. Repeater

Definition:
A repeater is a network device that receives a weak or degraded electrical signal and retransmits it at its original strength, allowing the signal to travel over longer distances.

Function:
A repeater operates at the Physical Layer (Layer 1) of the OSI model. As a signal travels over a cable, it weakens due to attenuation. The repeater amplifies and regenerates this signal so it can continue further without data loss. It does not filter or interpret data — it simply boosts the signal strength.

Figure: See diagram (Repeater — Weak Signal converted to Amplified Signal).

c. Hub

Definition:
A hub is a basic networking device that connects multiple computers or devices in a network and broadcasts incoming data to all connected devices regardless of the intended recipient.

Function:
A hub operates at the Physical Layer (Layer 1) of the OSI model. When a frame is received on one port, the hub forwards it to all other connected ports simultaneously. This causes unnecessary traffic and collisions, and hubs do not filter data or read MAC addresses. They have largely been replaced by switches in modern networks.

Figure: See diagram (Sender broadcasts to all Receivers through the Hub).

d. Switch

Definition:
A switch is an intelligent network device that connects multiple devices and forwards data only to the specific destination device, using MAC addresses.

Function:
A switch operates at the Data Link Layer (Layer 2) of the OSI model. It maintains a MAC address table that maps devices to ports. When a frame arrives, the switch checks the destination MAC address and forwards the frame only to the correct port, reducing unnecessary traffic and improving efficiency compared to a hub.

Figure: See diagram (PC 'A' sends a message only to PC 'F' through the Switch).

e. Bridge

Definition:
A bridge is a network device that connects two or more network segments and filters traffic between them based on MAC addresses.

Function:
A bridge operates at the Data Link Layer (Layer 2) of the OSI model. It divides a large network into smaller segments to reduce congestion, examining the destination MAC address of each incoming frame to decide whether to forward or block it across segments.

Figure: See diagram (Bridge connecting Segment 1 and Segment 2).

f. Router

Definition:
A router is a network device that forwards data packets between different networks by determining the best path using IP addresses.

Function:
A router operates at the Network Layer (Layer 3) of the OSI model. It connects multiple networks — for example, a home LAN to the Internet — and uses routing tables and protocols to determine the most efficient path for data. Routers commonly also provide NAT (Network Address Translation) and DHCP services.

Figure: See diagram (Internet → Modem → Router → Computer #1 and Computer #2).

g. Modem

Definition:
A modem (Modulator–Demodulator) is a device that converts digital signals from a computer into analog signals for transmission over telephone lines, and vice versa.

Function:
A modem operates at the Physical Layer (Layer 1) of the OSI model. On the sending side, it modulates digital data into analog signals suitable for the telephone network. On the receiving side, it demodulates the analog signal back into digital data the computer can use.

Figure: See diagram (digital signal → Modem → analog → Telephone Network → Modem → digital signal).

h. Network Interface Card (NIC)

Definition:
A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network, either through a wired (Ethernet) or wireless (Wi-Fi) connection.

Function:
A NIC operates at the Data Link Layer (Layer 2) of the OSI model. It converts data from the computer into a format suitable for network transmission, and vice versa. Each NIC has a unique MAC address burned into it, used to identify the device on the network.

Figure: See image (a physical Ethernet NIC card with an RJ-45 port).

i. Firewall

Definition:
A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules.

Function:
A firewall acts as a barrier between a trusted internal network and untrusted external networks such as the Internet. It inspects packets and allows or blocks them based on rules such as IP address, port number, and protocol, helping prevent unauthorized access, malware, and cyberattacks.

Figure: See diagram (Internet → Firewall → User 1, User 2, User 3).

j. Wireless Access Point (WAP)

Definition:
A Wireless Access Point (WAP) is a networking device that creates a wireless local area network (WLAN), allowing Wi-Fi enabled devices to connect to a wired network.

Function:
A WAP operates at the Data Link Layer (Layer 2) of the OSI model. It connects to a wired router or switch and broadcasts a Wi-Fi signal. Devices such as laptops and phones connect wirelessly to the WAP, which routes their traffic through the wired network. WAPs are commonly used to extend Wi-Fi coverage in large buildings or offices.

Figure: See diagram (ISP → Modem → Router → Switch → Wireless Access Point → Laptops).

k. VoIP Endpoint

Definition:
A VoIP (Voice over Internet Protocol) Endpoint is a device or application that sends and receives voice calls over an IP network instead of traditional telephone lines.

Function:
A VoIP endpoint converts voice audio into digital data packets and transmits them over the Internet. At the receiving end, the packets are reassembled in order and converted back into audio. Examples include IP phones, softphone apps, and video conferencing systems, which reduce communication costs and support voice, video, and messaging over a single network.

Figure: See diagram (IP Device Caller → Voice Audio Broken into Packets → Packets Sent Over Internet → Packets Reassembled and Played Back as Audio → Call Destination).

Conclusion

In this lab, we studied the fundamental network equipment used in modern computer networks. Each device serves a specific role at different layers of the OSI model. A repeater boosts weak signals; a hub broadcasts data to all connected devices; a switch intelligently forwards frames using MAC addresses; a bridge connects and filters traffic between network segments; a router connects different networks using IP addresses; a modem converts signals between digital and analog forms; a NIC provides the physical network connection with a unique MAC address; a firewall secures the network from unauthorized access; a WAP enables wireless connectivity; and a VoIP endpoint enables voice communication over IP networks. Understanding these devices is essential for designing, managing, and troubleshooting real-world computer networks efficiently and securely.